CRITICAL POINTS

Fermat's Theorem

The converse is false.

THE THEOREM SAYS

If f has a local maximum or minimum at c, then c is a critical point: $f'(c) = 0$ or $f'(c)$ does not exist.

IT DOES NOT SAY

~~If c is a critical point, then f has a local maximum or minimum at c.~~

READING THE THEOREM BACKWARDS

WRONG WORKING

$f(x) = x^3$ has $f'(0) = 0$, so $x = 0$ is a critical point, hence an extremum.

RIGHT WORKING

f increases through the origin and has no extremum there. A critical point is a candidate that must be tested further.

Candidates only: critical points are the only places a local extremum can occur; the derivative tests classify each one.